

DECLARATION OF USE OF GENERATIVE AI IN BACHELOR THESIS (TFG)

I have used Generative AI in this work

Check all that apply:

YES	NO
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If you have ticked YES, please complete the following 3 parts of this document:

Part 1: Reflection on ethical and responsible behaviour

Please be aware that the use of Generative AI carries some risks and may generate a series of consequences that affect the moral integrity of your performance with it. Therefore, we ask you to answer the following questions honestly (*please tick all that apply*):

Question		
1. In my interaction with Generative AI tools, I have submitted sensitive data with the consent of the data subjects.		
YES, I have used this data with permission	NO, I have used this data without authorisation	NO, I have not used sensitive data
2. In my interaction with Generative AI tools, I have submitted copyrighted materials with the permission of those concerned.		
YES, I have used these materials with permission	NO, I have used these materials without permission	NO, I have not used protected materials
3. In my interaction with Generative AI tools, I have submitted personal data with the consent of the data subjects.		
YES, I have used this data with permission	NO, I have used this data without authorisation	NO, I have not used personal data

4. My use of the Generative AI tool has **respected its terms of use**, as well as the essential ethical principles, not being maliciously oriented to obtain an inappropriate result for the work presented, that is to say, one that produces an impression or knowledge contrary to the reality of the results obtained, that supplants my own work or that could harm people.

YES

NO

If you **did NOT** have the permission of those concerned in any of questions 1, 2 or 3, briefly explain why (e.g. *"the materials were protected but permitted use for this purpose"* or *"the terms of use, which can be found at this address (...), prevent the use I have made, but it was essential given the nature of the work"*).

Not applicable.

Part 2: Declaration of technical use

Use the following model statement as many times as necessary, in order to reflect all types of iteration you have had with Generative AI tools. Include one example for each type of use where indicated: *[Add an example]*.

I declare that I have made use of the Generative AI system (Name of AI system/tool and version: ChatGPT, Gemini, Copilot...) **for:**

Documentation and drafting:

- *Supporting reflection in relation to the development of the work: iterative process of analysing alternatives and approaches using AI*

Not applicable

- *Revision or rewriting of previously drafted paragraphs*

Not applicable.

- *Search for information or answers to specific questions*

Not applicable.

- *Bibliography search*

I used Google Gemini to help me learn the BibTex format for citing sources in LaTeX. An example prompt would be: *Help me learn how to create BibTex entries. Here is one example of a book I have reviewed and want to cite as a source: Learn Algorithmic Trading Build and deploy algorithmic trading systems and strategies using Python and advanced data analysis, by Sebastien Donadio and Sourav Ghosh*

- *Summary of bibliography consulted*
- *Translation of texts consulted*

Develop specific content

Generative AI has been used as a support tool for the development of the specific content of the dissertation, including:

- *Assistance in the development of lines of code (programming)*

I have used Anthropic Claude via GitHub Copilot in the debugging and diagnosing of code mistakes. A particular example would be to diagnose a mistake I made with the Expectation Shortfall-like metric, where instead of measuring the 5% of worst errors I saved the 5% of best errors (smallest, not largest). This was a fine detail in the code which I would have probably taken a lot of time to debug, but it was easily diagnosed with the help of the LLM.

- *Generation of diagrams, images, audios or videos*

Not applicable, all graphs and diagrams are of own creation with either LaTeX or Matplotlib tools.

- *Optimisation processes*

Not applicable.

- *Data processing: collection, analysis, cross-checking of data...*

Not applicable, all of the data retrieval was done in Python, inspired by the knowledge acquired in “Arquitectura de Sistemas II”, a third-year subject where we developed a project interacting with the HTTP protocol in Java. The data processing and analysis techniques were learnt from the third-year subject “Modern theory of detection and estimation”, where we were taught Machine Learning workflows and data visualization skills.

- *Inspiration of ideas in the creative process*

Not applicable, the inspiration and direction of the research project was developed in conjunction with my advisor.

- *Other uses linked to the generation of specific points of the specific development of the work*

If you think it is necessary to include it, add it in each case:

using the *prompt* ...

(write the request made to the IAG)

having as interaction ...

(describe what interaction you made with the IAG after the prompt's response)

Example: *I declare that I have made use of the Generative AI system Chat GPT 3.5 to search for information using the prompt: "Tell me a concrete example that illustrates the application of the urban planning proposal of the 15-minute City in Spain", having as interaction the proposition of the concrete example of the Chamartín district that has been reflected in the report.*

Part 3: Reflection on utility

Please provide a personal assessment (free format) of the strengths and weaknesses you have identified in the use of Generative AI tools in the development of your work. Mention if it has helped you in the learning process, or in the development or drawing conclusions from your work.

My main use of LLMs in the research project was through the use of GitHub Copilot inside the code editor Visual Studio Code.

The general idea of the project was designed over pen and paper, during meetings with my advisor. The plan was over all straightforward, but implementing it at times resulted in mistakes that would lead to runtime errors, or logical mistakes in the calculations.

The use of LLMs was greatly beneficial in debugging my code and solving issues I had with the Python programming language. Even though it was learnt in "Modern theory of detection and estimation", and employed sparingly over other subjects, it was not the main programming language that was taught in my degree, which was Java.

In regard to its strengths, it helped me solve isolated mistakes that would have usually taken me much longer to pin-point inside my code. In regard to the weaknesses, at times it was very challenging to properly translate my issues to the LLM, as even though my plan seemed straight forward or logical to me, it not always understood my problems are they were not usually straightforward.

To conclude, it was greatly beneficial for my learning process in dealing with the Python language, and improving my knowledge in Machine Learning development. Using it as a coding companion, or how Microsoft calls it: "Copilot", has been extremely useful. Especially recalling how during my degree, I spent countless hours in Stack Overflow articles trying to solve particular Java or C issues, and you can now have your own personalized assistant that can tailor to your needs. It is an extremely powerful tool that has a lot of potential in the coming years, which must be used wisely, but it will definitely help us make a step forward in the technical industry. Whether it is by increasing the innovation or development speed of new systems and features, or by improving overall code quality assurance.